

Project Topic

Title: Scalable Real-Time Data Ingestion and Streaming Pipeline Architecture

In data-driven decision-making, the value of information decays rapidly over time, making traditional batch-processing methods obsolete for fast-paced industries. This project designs and implements a high-throughput, low-latency Real-Time Data Streaming Pipeline engineered to handle high-velocity data streams. Utilizing modern data engineering tools like Apache Kafka and Apache Spark, the system establishes distributed event brokers to ingest live data streams (such as user clickstreams, live financial markets, or IoT sensor logs). The pipeline performs on-the-fly streaming transformations—including data cleaning, schema validation, and aggregation—with sub-second latency. The final output is an automated data infrastructure that seamlessly moves raw data into a live data warehouse and updates a real-time analytics dashboard completely automatically.